ϕ^{-3} Gradient: Invertibility Test Results

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The Calculation

Tested whether the ϕ^{-3} confinement gradient transformation preserves particle identity.

Two models were run against known particle rung positions (normalized frame):

Particles tested: Electron (103.778), Muon (114.858), Proton (119.395), Tau (120.723), Z boson (128.906), Top quark (130.234)

Results

Model 1 — Linear Compression: $n_{final} = n \times \phi^{-3}$

Particle	n_initial	n_final	Recovered	Invertible
Electron	103.778	24.499	103.778	YES
Muon	114.858	27.114	114.858	YES
Proton	119.395	28.185	119.395	YES
Tau	120.723	28.499	120.723	YES
Z boson	128.906	30.431	128.906	YES
Top quark	130.234	30.744	130.234	YES

The integer rung identity is fully recoverable via ϕ^3 multiplication. The mapping is one-to-one.

Model 2 — Phase (Fractional Part) Preservation

Particle	frac_initial	frac_final	Preserved

Electron	0.778	0.499	NO
Muon	0.858	0.114	NO
Proton	0.395	0.185	NO
Tau	0.723	0.499	NO
Z boson	0.906	0.431	NO
Top quark	0.234	0.744	NO

The fractional phase position transforms under compression. Not preserved in original form.

Model 3 — Log Compression: $n\_final = \log\_\phi(n) \times \phi^{-3}$

Particle	n_initial	n_final	Unique mapping
Electron	103.778	2.277	YES
Muon	114.858	2.327	YES
Proton	119.395	2.346	YES
Tau	120.723	2.352	YES
Z boson	128.906	2.384	YES
Top quark	130.234	2.389	YES

Every particle maps to a distinct compressed value. One-to-one under log compression.

Summary

1. **Integer rung identity (n):** Survives ϕ^{-3} compression. Exactly recoverable via ϕ^3. The mapping is invertible.
2. **Fractional phase (resonance state):** Transforms under compression. Not preserved in original form. Recovery depends on whether the transformation function is known and deterministic at the ϕ^{-3} boundary.

3. **Many-to-one collapse:** Does not occur under either model. All particle identities remain distinguishable after compression.
